

1. Hyperparameters Grid Search Methods:

A.

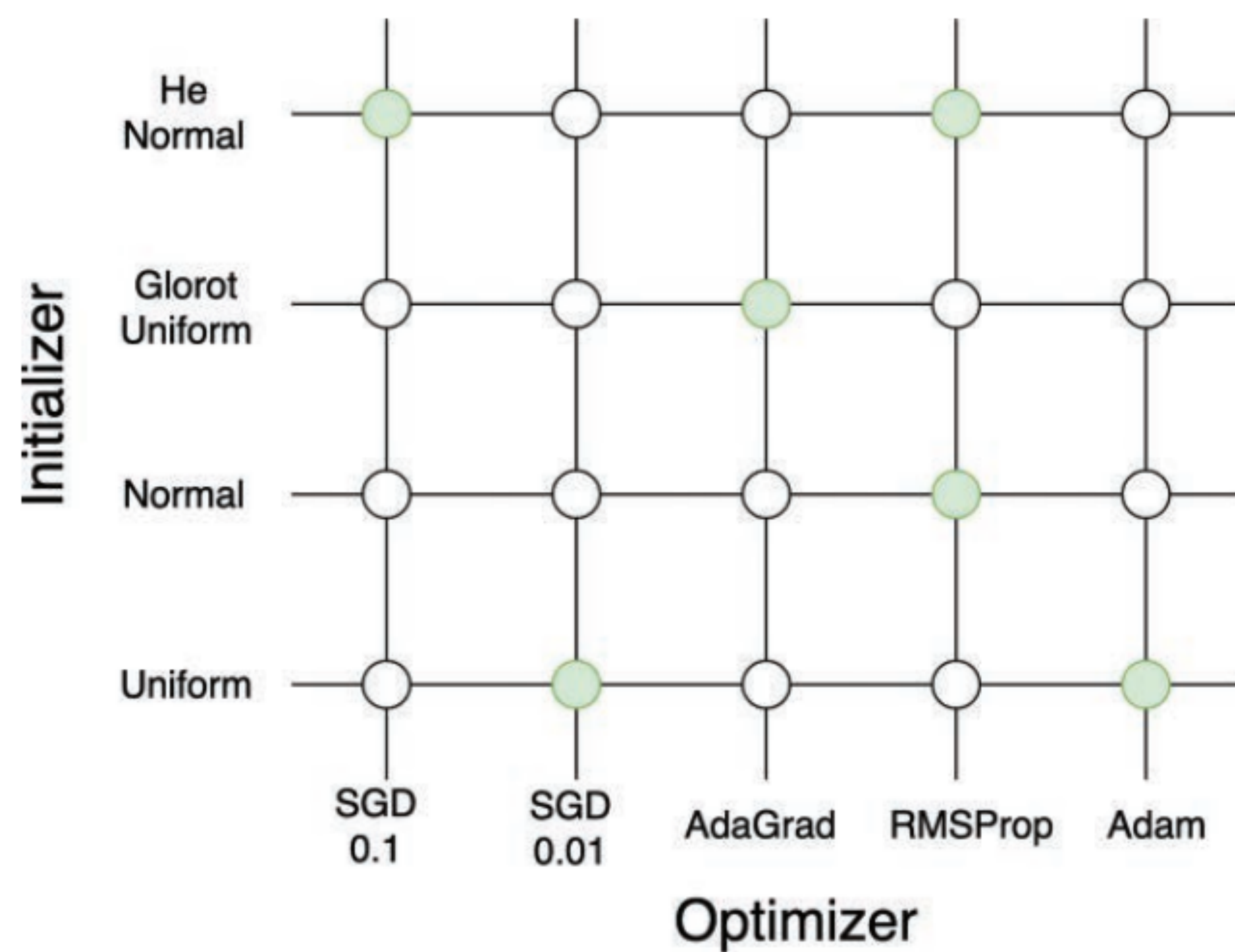


Figure 5-13 Grid search for two hyperparameters.

- B. hyperparameters grid search method — Exhaustive Grid Search: it evaluates every single combination across a multidimensional hyperparameter grid.
expensive and poorly scaling in computing all combinations as number of hyperparameter grows.
- C. hyperparameters grid search method — Random Grid Search: it evaluates random sets of all possible combinations, significantly cutting down computing time.
- D. hyperparameters grid search method — Hybrid Grid Search: it starts with the Random Grid Search to pinpoint the high potential regions in the grid, and then it performs the Exhaustive Grid Search around those specific regions in the grid.

2. Validation Dataset Importance:

A. tuning hyperparameters directly test datasets leads to severe data leakage, produced hyperparameters are apt to overfit the test datasets rather than generalize to unseen data.
by dividing whole data into three distinct datasets, we can solve overfitting.

B. three distinct datasets:

⇒ training dataset: it's used to tune parameters of neural networks and get **training error** during the training stage

⇒ validation dataset: it's used to tune hyperparameters of neural networks during the training stage

⇒ test dataset: it's used to evaluate how well neural networks are able to generalize to previously unseen data get **test error** during the training stage

C. standard split ratio of three distinct datasets is 60 / 20 / 20.

3. k-Fold Cross Validation:

A. we need to find out a way to overcome the limitation where a static population of the validation dataset reduces volume of the training dataset to only 60% of the whole dataset.

B. k-Fold Cross Validation procedure:

⇒ split the whole dataset into the 80% training dataset and the 20% test dataset.

⇒ divide the 80% training dataset into k parts (ex: $k=5$ folds).

⇒ allocate $k-1$ folds for the training dataset and 1 fold for the validation dataset, which create k unique instances for k separate runs in the training stage.

For $k = 5$, this yields 5 unique instances trained from scratch:

- Run 1: Folds 1, 2, 3, 4 (train) + Fold 5 (val)
- Run 2: Folds 1, 2, 3, 5 (train) + Fold 4 (val)
- Run 3: Folds 1, 2, 4, 5 (train) + Fold 3 (val)
- Run 4: Folds 1, 3, 4, 5 (train) + Fold 2 (val)
- Run 5: Folds 2, 3, 4, 5 (train) + Fold 1 (val)

⇒ average the validation scores across all k times runs to establish the final score for hyperparameters tuning (= hyperparameter combinations selecting).

⇒ once the best hyperparameters (= hyperparameter combinations) configuration is identified,

we retrain the neural network by using the k folds dataset (= the 80% training dataset) as the training dataset, and evaluate the final neural network by utilizing the 20% test dataset.